

ICRS Robotics 101

Session 2 (Electrical)

Does your thing need electronics?

- Whenever youre building something, first chart out any general electrical requirements
- Sensing
- Actuating
- Display

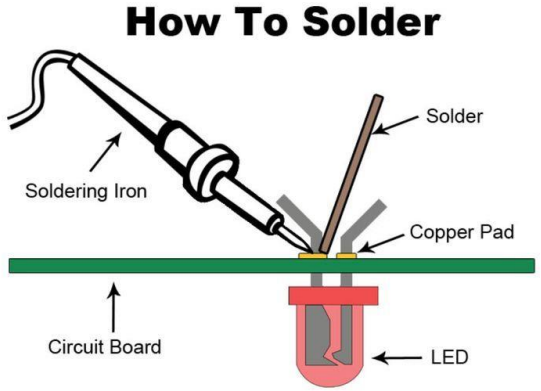
Electrical Design

- Keep it simple
- Popular components are better
- You will break things very often
- You can fix almost every hardware issue in software

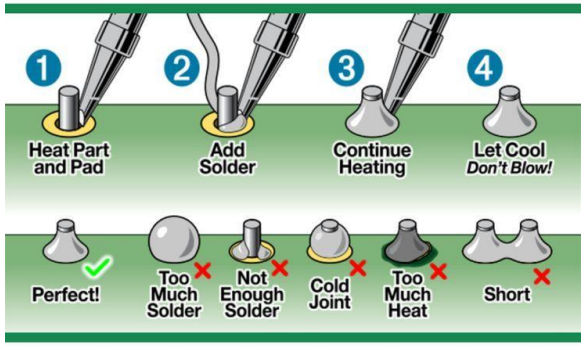
General electrical tips

- Try make a basic diagram to have something to reference
- If something is hot or smoking, unplug it
- Breadboards are good for beginners but get limiting quickly
- Learn what signals you should be expecting and where

Soldering

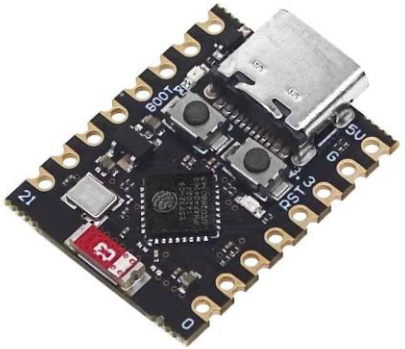


- Most important electronics skill
- Soldering iron tip should be shiny



- Its easier to add than remove solder
- Before turning it on, check for shorts with multimeter

Microcontroller (MCU)

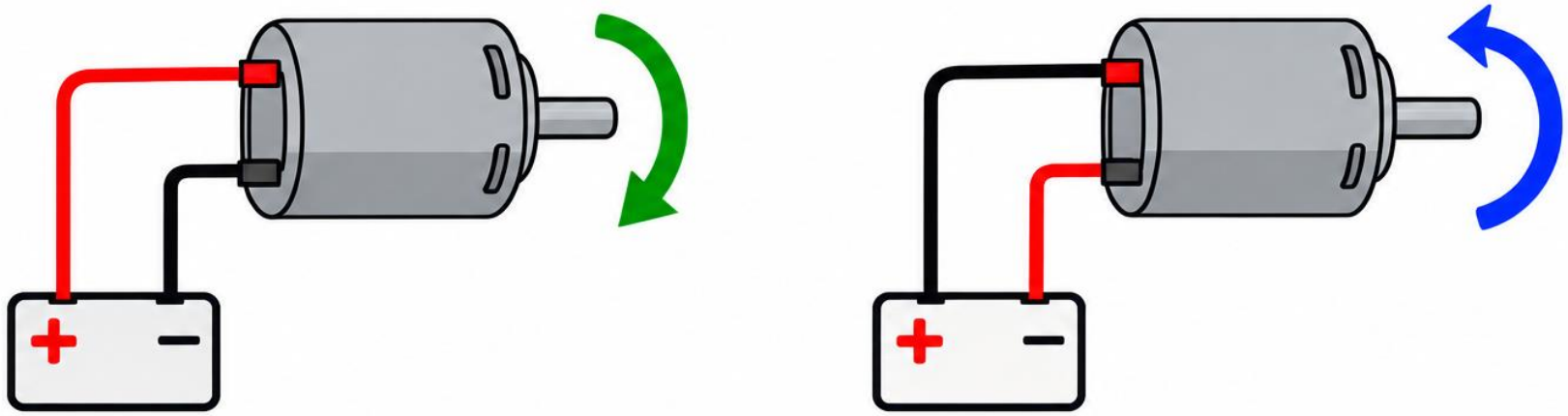


- ESP32 C3 SuperMini Development Board
- Brain of the robot, where you put the code
- Embedded coding is very different to python coding (in a fun way)
- It interacts with the world via digital and analogue pins
- Its either high/low or talking to something over a communication protocol

N20 Mini Micro Metal Gear Motor (DC Motor)

- First example of an actuator

- DC motors just need power and ground and then spin

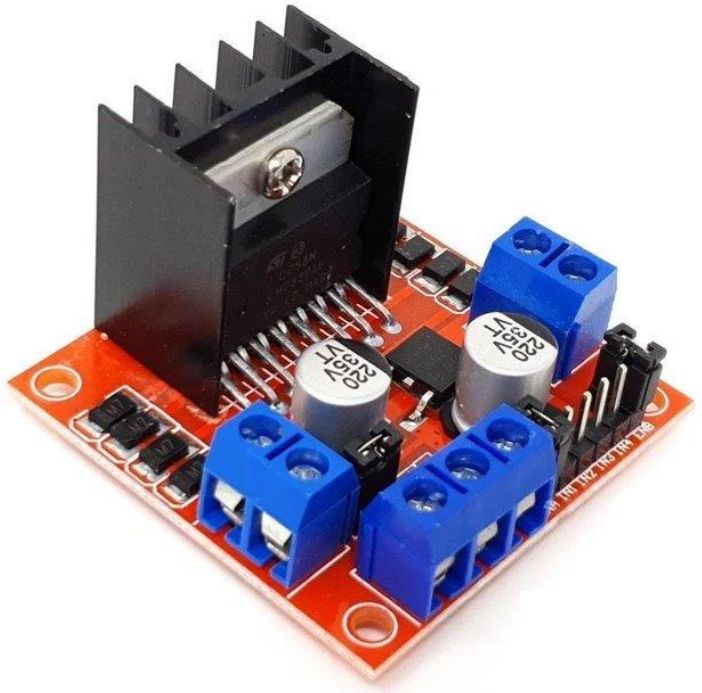


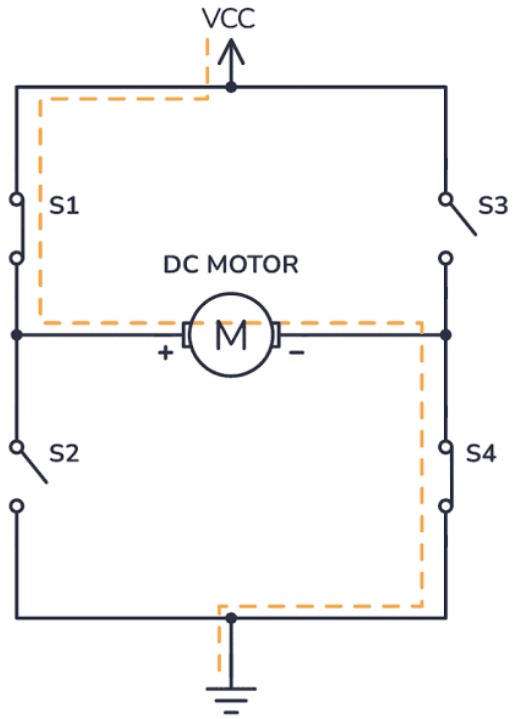
Logic Power and Motor Power

- 2 power regimes
- Logic power carries sensitive signals that don’t like noise
- Motor power needs higher currents and higher voltages
- On our robot that will be 9V and 3.3V (usually written 3V3)

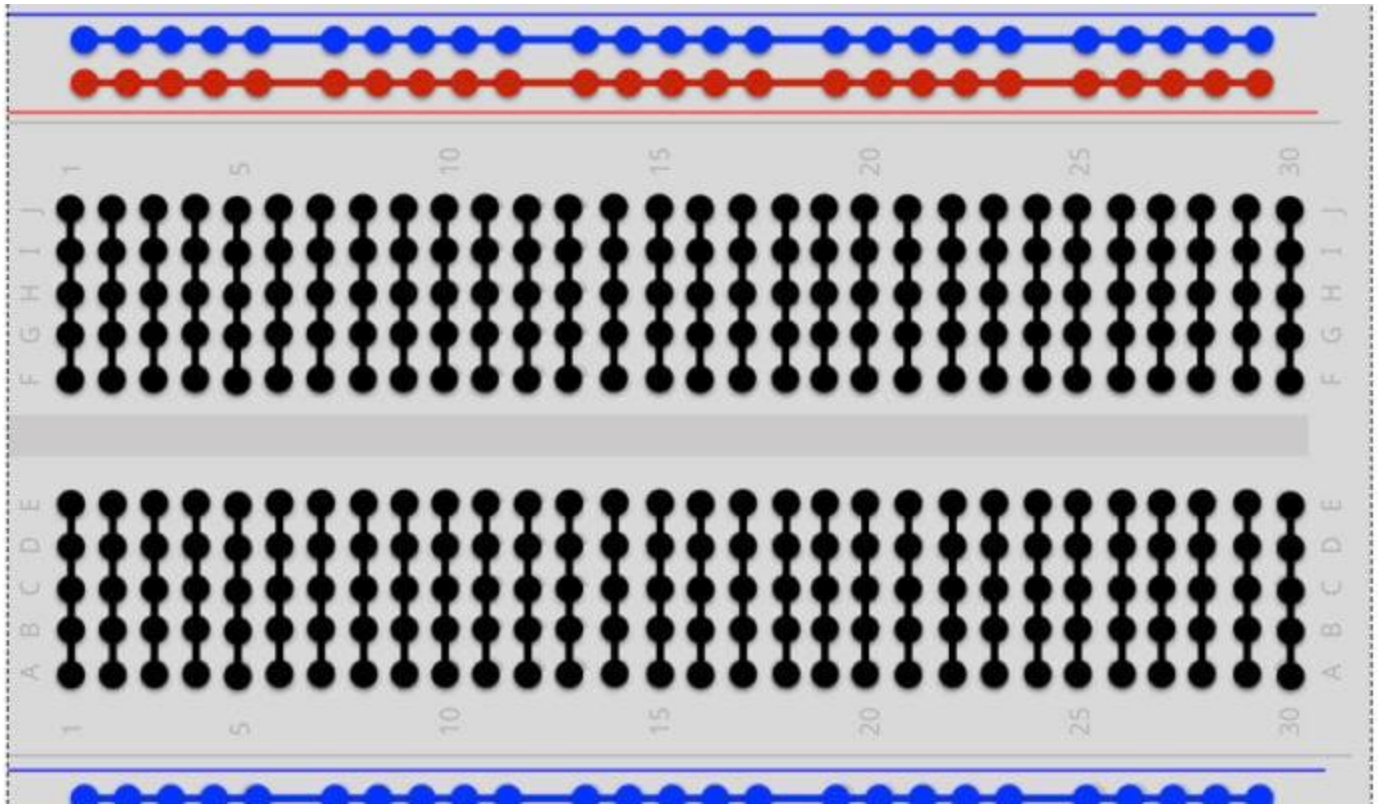
L298N (Motor Driver)

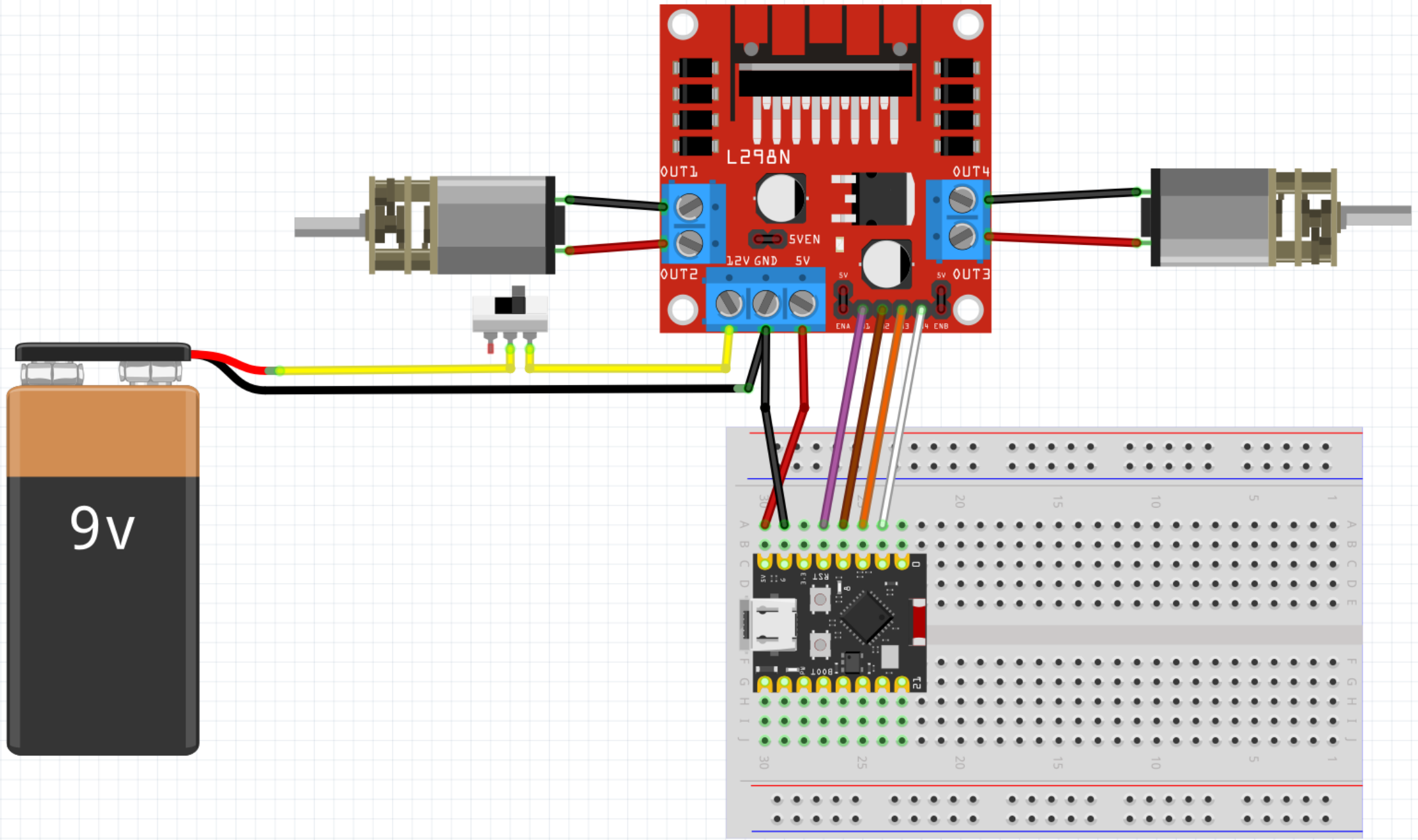
- Drives the motor
- Converts power from a battery into power that can make a motor spin
- Has signal pins that the MCU can talk to





Breadboards



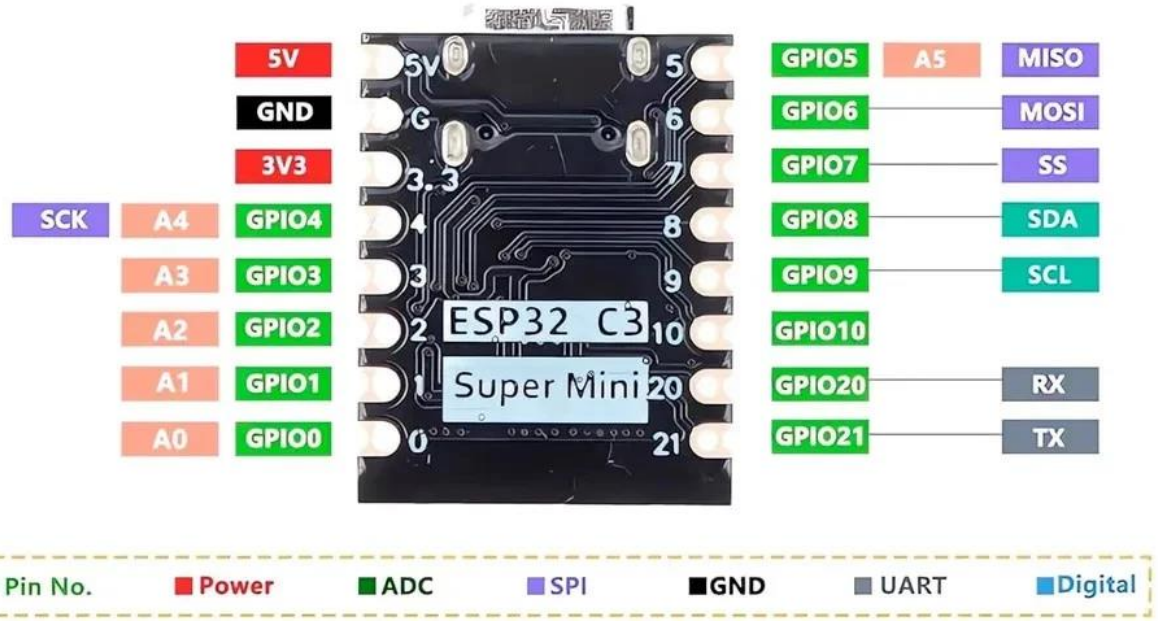

L298N
OUT1
OUT2
OUT3
OUT4
5VEN
12V GND 5V
5V
5V
ENA
ENB
9v

Sensible assembly order

- Cut the red wire on the battery connector and solder in the switch
- Solder on the pins to the ESP32 and put it on the breadboard
- Connect power up AND THEN turn the battery on. Swapping PWR and GND can break the ESP32
- Solder wires onto the motor
- Connect the control pins using jumper wires

DC – DC Power conversion

- So, we have only a 9V battery powering the device so where is 3V3 coming from?
- ESP32 can "Buck" down 5V to 3V3
- L298N can "Buck" down 12V to 5V
- Going the other way is "Boosting" and a "Buck-Boost" can do both ways


5V
GND
3V3
SCK
A4
GPIO4
A3
GPIO3
A2
GPIO2
A1
GPIO1
A0
GPIO0
ESP32 C3
Super Mini
GPIO5
A5
MISO
GPIO6
MOSI
GPIO7
SS
GPIO8
SDA
GPIO9
SCL
GPIO10
GPIO20
RX
GPIO21
TX
Pin No.
Power
ADC
SPI
GND
UART
Digital



